

# Economic-Aware and Sovereignty-Constrained Routing for Enterprise LLM Gateways

**Draft v0.2 — single-provider (OpenAI) evaluation.** A first complete write-up built from real

measurements

## Abstract

Enterprises are adopting large language models (LLMs) faster than they can govern them: they cannot

say which model,

## 1. Introduction

LLMs are now embedded in HR assistants, document RAG, developer tooling, and analytical agents. Each

call has a

1. **Cost opacity.** Teams cannot attribute spend per application/team or decide when a cheaper model

would suffice

Existing gateways and FinOps tools mostly \*measure and report\*. Our thesis is that a control plane at

the gateway

## Contributions.

- A cons

## 2. Background and Related Work

References are in `references.bib` (recent, verified); citation keys are shown in brackets.

**Model routing and cascades.** A growing line of work trades quality for cost by selecting among

models per  
a

**LLM serving systems.** Orca [yu2022orca], vLLM/PagedAttention [kwon2023pagedattention], SGLang

[zheng2023

**LLM gateways.** Envoy AI Gateway [envoyaigateway] and LiteLLM [litellm] provide a unified data path

to multiple

**Evaluation.** We follow the LLM-as-a-judge methodology of Zheng et al. [zheng2023judge] (MT-Bench /

Chatbot A  
we adopt

**Regulation.** Data-residency and sensitive-data handling are driven by the GDPR [gdpr2016] and the

EU AI Act [

## 3. Problem Formulation and Algorithm

A request  $r$  carries metadata (team, namespace, sensitivity) and an estimated token footprint. Given

a catalog

**Hard constraint.** Reject any  $m$  that violates  $P$ : its zone is forbidden; or an allow-list is set

and  $m$ 's

**Soft objective.** Among  $C$  meeting a minimum quality  $q_{\min}$  (relaxed only when the budget is

exhausted

$$\arg\min_{m \in C} \alpha \cdot \widehat{c}(m) \cdot (1 + \epsilon \cdot \rho) + \beta \cdot \ell_q(m) + \gamma \cdot \ell_\lambda(m) + \delta \cdot \sigma(m)$$

where  $\widehat{c}(m) = \frac{\text{cost}(m)}{\text{cost}(\text{premium})}$  is normalized cost,  $\rho = \text{used}/\text{total}$

is budget  
expected

Pseudocode and the exact normalization are in `methodology.md`; the implementation is

`experime

## 4. System Design

Requests flow client → **Envoy AI Gateway** → provider (managed API or self-hosted). The **AI**

Sovereign  
**budgets,**

## 5. Methodology

**Testbed.** A Go harness drives the operator's *actual* engines and issues **real** LLM calls. The

provider in

**Workloads (W1-W4).** HR chatbot (short, sensitive), RAG docs (long context, quality-sensitive), Dev

assistant (

**Baselines.** B1 premium-static, B2 round-robin, B3 least-cost, B4 static namespace policy, B5 budget

hard-block

**Metrics.** Cost (total, per request, per token, per team via the operator cost engine, % savings);

quality (LL  
the

**Reproducibility.** Temperature 0; all responses and judge scores cached (`results/cache.json`) so

re-runs are

## 6. Results

### RQ1 — Cost (Figure 2, Table 1)

Relative to

### RQ2 — Quality preservation (Figure 3, Table 2)

**B6-ours**  
97.5%),

### RQ3 — Latency and overhead (Figure 4, Table 3)

The **routing**

### RQ4 — Sovereignty (Figure 5, Table 4)

A sovereig

### RQ5 — Budget-aware degradation (Figure 6, Table 5)

Under a b  
**availabili**

### RQ6 — Managed vs self-hosted break-even (Figure 7) — *\*modeled\**

Using rea  
managed

### Ablation (Figure 8, Table 7)

Removing  
term

## 7. Discussion

Economic-aware routing is most valuable when a workload mixes easy and hard requests: the router sends

cheap req

## 8. Threats to Validity

Summarized here; full version in `threats\_to\_validity.md`. **Internal:** API/latency variability

(mitigated

## 9. Conclusion and Future Work

A gateway-level, economic-aware, sovereignty-constrained, budget-aware control plane can cut LLM cost

substantia

## Artifact / Reproducibility

Operator + `experimentation/` (datasets, harness, scripts, results, figures, cached responses).

Reproduce

## References

See `references.bib` (BibTeX). Key works cited: model routing/cascades — FrugalGPT

[chen2023

## Appendix: Figures

Figure 2 — Total cost by routing strategy

0.040  
0.035  
0.030  
0.025  
0.020  
0.015  
0.010  
0.005  
0.000

Total cost (EUR)

Figure 3 — Quality vs cost

0.90  
0.89  
0.88  
0.87  
0.86

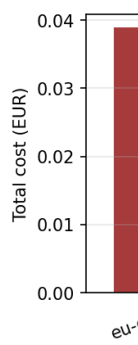
Mean quality (normalized 0-1)

**Figure 4 — Latency p95 and mean  $\pm$ 95% CI**

3000 -  
2500 -  
2000 -  
1500 -  
1000 -  
500 -  
0 -

Latency (ms)

**Figure 5 — Sovereignty: cost & violations**

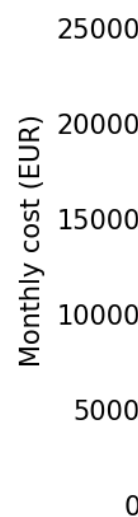


**Figure 6 — Budget policy: availability vs overrun**



**Figure 7 — Managed vs self-hosted break-even (modeled)**





**Figure 8 — Ablation of scoring terms**

